

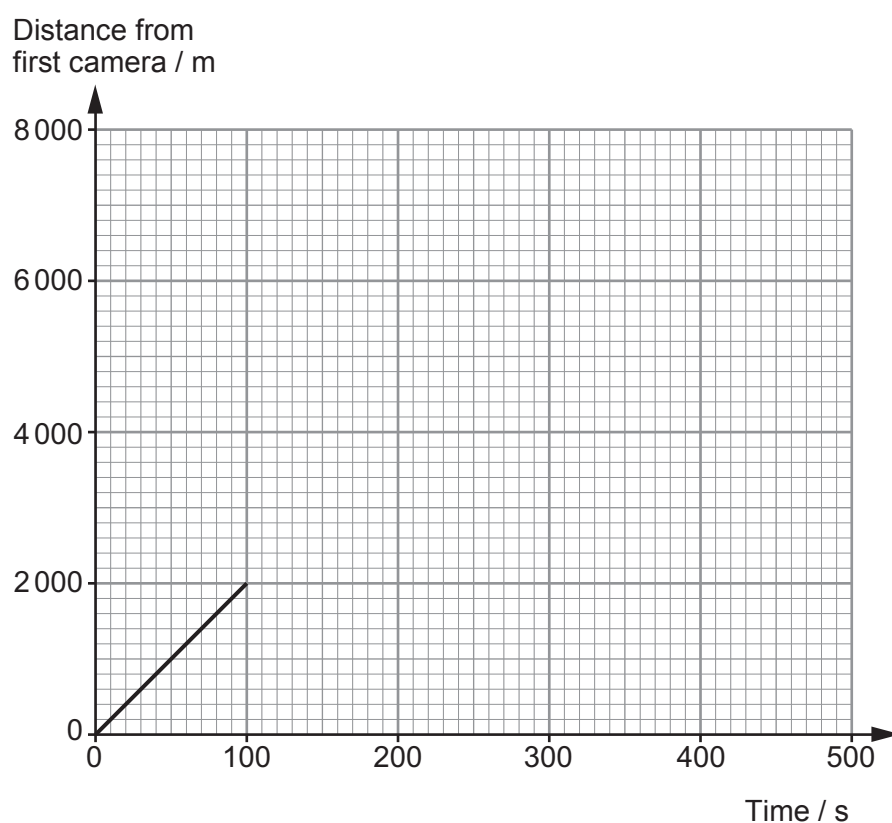
2. (a) State the difference between a vector and a scalar and give one example of each. [2]

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- (b) Average (or mean) speed areas are now found on many motorways. A section of motorway has a mean speed of 60 km h^{-1} . The mean speed is monitored between two cameras placed 8 km apart. The graph shows the first 100 seconds of the motion of a car as it travels between the two cameras.



- (i) Show that the instantaneous speed of the car at $t = 50\text{ s}$ is approximately 70 km h^{-1} . [2]

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- (ii) The car covers the 8 km between the two cameras at a mean speed of exactly 60 km h^{-1} . Complete the graph to show this motion. [3]
Space is provided below for your calculations.

- (c) On another stretch of motorway, the speed limit is equivalent to 30 ms^{-1} . The maximum deceleration a normal car can achieve in dry conditions is assumed to be 8 ms^{-2} . Traffic police can estimate the speed of vehicles involved in accidents using the length of the marks made by skidding tyres on the road. In one accident, the skid marks were found to be 85 m long. Investigate whether or not the car that created the skid marks was travelling below the speed limit prior to the accident. [3]

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